

Unit 1

Introduction to Research

Research is something we all do, even if we don't realise it. Whenever we try to find out something new, look for an answer to a question, or try to understand why something happens, we are doing research. It's not just for scientists or scholars—it's part of everyday life. But in formal education and professional fields, research has a more specific meaning. In this module, we'll explore what research really means, why it matters, how it's done, and what makes it good or bad.

What is Research?

Research is a careful and organised way of finding answers to questions. It is about collecting facts, examining them, and then coming up with a conclusion based on that evidence. It's not guessing, assuming, or relying on opinions. Instead, it's about using a methodical approach to understand a topic better.

Let's say a doctor wants to find out whether a new medicine works better than the old one. They can't just try it on a few people and guess. They need a plan, a method, a process—and that's research.

Research can be applied in science, medicine, business, education, social studies, engineering, and almost every field. It helps solve problems, build knowledge, and improve the way things are done.

Objectives of Research

There are different reasons why people carry out research. These are known as the objectives of research. Some common objectives include:

1. To discover new facts – Learning something that wasn't known before.
2. To verify and test existing facts – Checking if what we already believe is actually true.
3. To analyse relationships between different variables – For example, does studying more hours lead to better marks?

4. To find solutions to problems – Whether it's a social issue, a health issue, or a technical problem, research helps find practical answers.
5. To develop new tools or systems – Research often leads to the creation of new technologies, teaching methods, machines, or procedures.
6. To help in decision-making – Governments, companies, and schools use research to make better choices.

So, the main purpose of research is not just about gathering information, but making sense of it and using it to do something meaningful.

Types of Research

Not all research is the same. Depending on the goal, the subject, and the method used, research can be divided into different types. Here are some of the main ones:

1. **Basic Research:** Also called pure research, this type of research is done to expand our knowledge. It doesn't always have a practical use at the moment, but it helps us understand how things work. For example, studying how memory works in the human brain.
2. **Applied Research:** This research is done to solve real-world problems. It is more practical. For example, developing a new vaccine or improving farming methods.
3. **Descriptive Research:** This type aims to describe things as they are. It answers questions like "what is happening?" or "what do people think?". Surveys and case studies often fall under this.
4. **Analytical Research:** This type tries to understand why something is happening. It looks at the data and analyses it deeply.
5. **Quantitative Research:** This involves numbers and measurements. It is used when we need to collect data that can be counted or measured, like how many students pass a test.
6. **Qualitative Research:** This is more about understanding feelings, thoughts, and experiences. It doesn't rely on numbers but on interviews, observations, and open-ended questions.

Research Approaches

When we talk about research approaches, we mean the overall plan or way in which the research is carried out. The two most common approaches are:

1. Quantitative Approach

- Based on numbers and data.
- It is structured and uses tools like surveys, tests, or experiments.
- Useful for studies that need results in measurable form.

2. Qualitative Approach

- Based on opinions, experiences, and meanings.
- It's more flexible and uses interviews, observations, or case studies.
- Ideal for understanding deep insights or behaviours.

Sometimes, both approaches are used together, known as the **Mixed Method Approach**. This can give a fuller understanding of a problem by combining numbers with human insights.

Research Methods vs Methodology

This is a topic many students find confusing, so let's break it down as simply as possible.

What are Research Methods?

Research methods are the **tools** or **techniques** used to carry out research. These are the practical steps you take to collect data or information. Think of them as the “how” part of research.

For example:

- Doing a **survey** to find out what people think.
- Having **interviews** with a group of students.
- Carrying out an **experiment** to test how plants grow under different lights.
- Watching and writing down what people do in a particular setting (**observation**).

These are all research methods because they help you gather information to answer your research question.

What is Research Methodology?

Now, methodology is a bit different. It's not about the tools, but the thinking behind choosing those tools. Methodology is the study of how research should be done. It explains why a particular method is suitable for your topic and how it fits into the bigger picture of your study.

So, if methods are the tools, methodology is the reasoning behind using those tools.

Example to understand the difference:

Let's say you're doing research on how students feel about online classes.

- Your method might be: an online questionnaire with 10 questions.
- Your methodology would explain:
 - Why you chose a questionnaire.
 - Why online was better than face-to-face in your situation.
 - Why your questions were open-ended instead of multiple-choice.

In simple words:

- Methods = What you did
- Methodology = Why you did it that way

Research and the Scientific Method

The scientific method is a systematic way of thinking and doing research, especially in science and technical subjects. But the steps can be used in any research to make it reliable and logical.

Let's break it down:

Step 1: Ask a Question / Identify the Problem: All research starts with a question. This could be something like, "Why do students perform better when studying in groups?"

Step 2: Do Background Research: You look up what others have already found out about this topic. You read books, articles, or previous studies. This helps you understand the topic better and avoid repeating work already done.

Step 3: Form a Hypothesis: A hypothesis is an educated guess. It's what you think might be the answer based on your background research.

Example: "Students who study in groups remember more than those who study alone."

Step 4: Test the Hypothesis (Experiment or Observation): Now you try to find out if your guess was right. You might run a test, do a survey, or observe behaviour.

You gather facts to support or reject your hypothesis.

Step 5: Analyse the Data: Once you've collected your data, you need to make sense of it.

- Did it support your hypothesis?
- Were the results surprising?
- Did different students react differently?

Step 6: Draw a Conclusion: Based on your analysis, you decide whether your hypothesis was correct or not. You also explain what the results mean in real life.

Step 7: Report or Share the Results

In proper research, it's important to share findings, so others can learn or build on your work.

Why is this method useful?

- It's clear and step-by-step.
- It avoids guesswork.
- It's based on evidence, not just opinion.
- It can be repeated by others.

Research Process

The research process is a series of steps followed to complete a research study properly. No matter what subject you're working in, these steps help make your work clear, honest, and useful.

Here's the process explained simply:

1. Choosing the Topic: Start with a topic you're interested in. It should be something you can explore in detail and is not too broad or too narrow.

Example: "Impact of social media on teenagers' sleep."

2. Review of Literature: This means reading what others have already said or discovered about your topic.

It helps you:

- Avoid repeating existing research.
- Build on past knowledge.
- Find gaps in the research.

3. Forming a Research Question or Hypothesis: Your research needs a clear direction. A research question could be: "Does using social media at night affect sleep duration?"

A hypothesis could be: "Teenagers who use social media after 10pm sleep less than 6 hours."

4. Selecting the Research Method: Here, you decide how to collect your data. Options include:

- Survey
- Interview
- Observation
- Experiment

5. Collecting Data: Now you go out and actually gather your information. For example, giving out questionnaires or recording results of an experiment.

6. Analysing the Data

This means looking at the data carefully.

- If it's numbers, you might make graphs or tables.
- If it's opinions, you might look for common themes.

7. Drawing Conclusions: Based on your data, what did you find out?

- Did your research question get answered?
- Was your hypothesis correct?

8. Writing the Report / Presentation

Finally, you write everything down in a clear way. This includes:

- What you studied
- How you did it
- What you found
- What it means

You can share this as a report, PowerPoint presentation, or even publish it, depending on your field.

Characteristics of Research: Good research isn't just any study. It has special features that make it trustworthy and useful. Let's look at these characteristics one by one, in a simple way:

1. **Systematic:** It follows a clear plan or order. It's not random or rushed.
2. **Logical:** Every step is based on reason. It must make sense and follow a proper sequence.
3. **Objective:** The research should not be biased. The results should not depend on what the researcher personally wants to find.
4. **Evidence-based (Empirical):** It is based on real data or observation, not just opinions.
5. **Replicable:** Others should be able to do the same research and get similar results. This makes it reliable.
6. **Clear and Simple:** The language and explanation should be easy to understand, especially for those who are not experts.
7. **Ethical:** It respects the rights, privacy, and dignity of people involved. It doesn't cause harm.
8. **Accurate:** Facts and figures must be correct. Mistakes can ruin the research.
9. **Useful:** Good research helps solve a problem, improve a situation, or add something new to what we know.

Criteria of Good Research

Now let's understand what makes research "good". Not all research is of high quality. These points help us judge whether a research study is worth trusting.

1. **Well-defined Purpose:** The aim of the research must be clear. You should know what the researcher is trying to find out from the beginning.
2. **Detailed Planning:** A good research study is carefully planned before it starts. Every step should be thought through to avoid confusion later.
3. **High Ethical Standards:** Researchers must not harm or cheat the people involved. They must get permission (called informed consent) and protect their privacy.
4. **Accurate and Honest Data:** The data must be recorded truthfully. No faking of numbers, no leaving out details just to get better results.
5. **Proper Tools and Techniques:** The researcher must use methods that suit the topic. For example, don't use a Maths-based survey to study feelings.
6. **Sound Analysis:** The data must be studied properly. Mistakes in analysis can give wrong results.
7. **Unbiased Conclusion:** Don't try to force the conclusion to match what you wanted to find. Just report what the data actually says.
8. **Clear Presentation:** The final report or presentation should be simple, clear, and well-organised so others can understand and learn from it.